THE LOWER SHRIEK FUNCTOR IS INDEPENDENT OF THE CHOSEN COMPACTIFICATION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0G50 (sources/stacks-duality/pinned-original-source.tex line 8688).

Allowed standard prerequisites (editorial).

- Cofilteredness of the category of compactifications. More on Flatness, Theorem labelled flat-theorem-nagata and Lemmas labelled flat-lemma-compactifications-cofiltered and flat-lemma-compactifyable, cited at proof lines 8700-8706. Allows any two compactifications to be compared.
- Compatibility of the extension by zero with the morphism of compactifications. Same chapter, Lemma labelled lemma-well-defined, cited at proof lines 8712-8716. Gives the displayed isomorphism.

Source reference. Original theorem 0F41: sources/stacks-chow/tags.txt, source lines 18886–18886. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 8699 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0ATU: sources/stacks-chow/tags.txt, source lines 13627–13627. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 8700 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0A9Z: sources/stacks-chow/tags.txt, source lines 12967–12967. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 8701 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Date: stacks-project:duality.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0G4X: sources/stacks-duality/pinned-original-source.tex, source lines 8577–8577. Same-chapter pointer into the author environment lemma at source line 8577; invoked at proof line 8715 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *The functor $Rf_!$ is, up to isomorphism, independent of the choice of the compactification.*

Proof. Consider the category of compactifications of X over Y, which is cofiltered according to More on Flatness, Theorem 0F41 and Lemmas 0ATU and 0A9Z. To every choice of a compactification

$$j : X \to \overline{X}, \quad \overline{f} : \overline{X} \to Y$$

the construction above associates the functor $R\overline{f}_* \circ Rj_!$. Suppose given a morphism $g : \overline{X}_1 \to \overline{X}_2$ between compactifications $j_i : X \to \overline{X}_i$ over Y. Then we get an isomorphism

$$R\overline{f}_{2,*} \circ Rj_{2,!} = R\overline{f}_{2,*} \circ Rg_* \circ j_{1,!} = R\overline{f}_{1,*} \circ Rj_{1,!}$$

using Lemma 0G4X in the first equality. In this way we see our functor is independent of the choice of compactification up to isomorphism. □